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Data Article

Lentil plant disease and quality assessment: A detailed dataset of high-resolution images for deep learning research

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A R T I C L E I N F O

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A B S T R A C T

The Lentil, a vital legume globally cultivated, faces significant challenges from diseases like ascochyta blight, lentil rust, and powdery mildew. Ensuring optimal harvest timing and effectively discerning healthy and diseased lentil plants are crucial for maintaining crop quality and economic viability, particularly in regions such as Bangladesh. This paper introduces a comprehensive dataset comprising high-resolution images of lentil plants gathered meticulously over four months from diverse locations across Bangladesh, under expert supervision. The dataset aims to support the development of machine-learning models for precise disease detection and quality assessment in lentil cultivation. Potential applications include enhancing the accuracy of quality evaluation, and improving packaging processes, thereby enhancing overall lentil production efficiency. Agricultural researchers can utilize this dataset to advance applications of computer vision and deep learning in managing crop diseases and enhancing yield outcomes. The dataset's creation involved collaboration with domain experts to ensure its relevance and reliability for agricultural research. By leveraging this dataset, researchers can explore innovative approaches to tackle challenges in lentil farming, contributing to sustainable agricultural practices and food security. Moreover, the dataset serves as a valuable resource for training and testing machine learning algorithms tailored to agricultural settings, facilitating advancements in automated agricultural technologies. Ultimately, this initia-

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tive aims to empower stakeholders in the lentil industry with tools to mitigate disease impact and optimize production practices, paving the way for more resilient and efficient agricultural systems globally.

Specifications Table

Subject	[Computer Science]
Specific subject area	[Lentil Disease Diagnosis, Agricultural Imaging, Crop Health Monitoring, Image Recognition, Deep Learning, and Computer Vision]
Type of data	[Image]
Data collection	[This dataset contains images representing different stages and conditions of lentil crops, including healthy plants and those afflicted by diseases such as ascochyta blight, lentil rust, and powdery mildew. The images were carefully collected between May and August 2023 from farms in Barisal, Bangladesh, using a Samsung M31 smartphone. Each image is sized at 224 × 224 pixels, stored in JPG format, and meticulously labeled by disease class to facilitate accurate classification and analysis. This resource is crucial for advancing plant pathology research and developing AI-driven disease detection tools. Special thanks to Mohammad Enayet-e-Rabbi, Deputy Director of Quality Control at the Seed Certification Agency, Ministry of Agriculture, Bangladesh, for his invaluable feedback and cooperation.]
Data source location	Location: [Various farms and research stations in Barisal, Bangladesh Coordinates: 1. Farm A: 22.7083° N, 90.3653° E 2. Farm B: 22.6715° N, 90.3252° E 3. Research Station C: 22.7032° N, 90.3863° E Zone: Barisal] Country: [Bangladesh]
Data accessibility	Repository name: [Mendeley Data] Data identification number: 10.17632/7vb77bz2st.1 Direct URL to data: https://data.mendeley.com/datasets/7vb77bz2st/1
Related research article	[None]
How the dataset helps in packaging	1. [**Quality Assessment**: Automates the assessment of lentil quality, ensuring only high-quality products are packaged. 2. **Sorting and Grading**: Aids in developing algorithms for sorting lentils based on size, color, and disease presence, enhancing efficiency. 3. **Disease Detection**: Helps in developing systems that detect and remove diseased lentils during packaging. 4. **Market Readiness**: Provides insights into consumer preferences for lentil appearance, informing packaging design. 5. **Sustainable Practices**: Reduces waste through effective sorting, aligning with sustainable farming trends. 6. **Integration with Robotics**: Supports the development of robotic packaging systems that identify optimal packing methods based on lentil characteristics.]

1. Value of the Data

- Critical for Agriculture and Economy: Lentils are a staple crop in Bangladesh, significantly contributing to the agricultural economy. Inconsistent human inspection and misidentification of diseases can lead to reduced yields and financial losses. This dataset enables non-invasive image analysis for accurate disease classification, facilitating the development of automated systems that provide precise disease management recommendations. By enhancing [1] crop yields and reducing dependency on imports, this dataset strengthens the nation's food security, which is vital for Bangladesh, where agriculture is a primary livelihood source.

- Boosting Farmers' Income and Food Security: Early and accurate detection of lentil diseases is essential for maintaining crop quality and ensuring farmers' economic stability [2]. This dataset helps prevent crop losses, securing farmers' income and reducing poverty, particularly for smallholder farmers. Crops play a vital role in driving economic growth and fostering development.
- Benefits for the General Population: The benefits of this dataset extend beyond farmers to the broader population. Healthier crops result in a stable supply of lentils, a key protein source in the Bangladeshi diet. Improved crop yields and minimized waste can stabilize or reduce food prices, making nutritious food more accessible. Better disease management may also decrease the reliance on chemical pesticides, resulting in healthier food products.
- Driving Research and Technological Innovation: This dataset serves as an invaluable resource for researchers focusing on computer vision, machine learning, and agricultural sciences. It encourages multidisciplinary work by providing a solid basis for creating automated systems to forecast and control crop health. Sustainable agricultural development in Bangladesh is supported by the dataset, which improves crop quality, decreases labour costs, and boosts efficiency in disease management [3].
- Empowering Researchers and Advancing Knowledge: Availability of this dataset presents a unique opportunity for researchers to engage in studies bridging technology and agriculture. It serves as a critical resource for testing algorithms and refining models in real-world scenarios. The dataset opens doors for collaboration with international research institutions, enabling Bangladeshi researchers to contribute to global advancements in agricultural technology and publish insights in high-impact journals, elevating the profile of Bangladesh's research community.

2. Background

The inspiration for creating this dataset stems from the significant challenges faced in identifying lentil diseases within the agricultural sector. Precision agriculture, which focuses on enhancing crop management through technological innovations, has long needed a comprehensive dataset specific to lentil diseases. Without such data, developing accurate detection models has been a struggle. This dataset, with its 1,612 images of lentil crops showing various disease stages, is a crucial resource for training and validating deep learning algorithms. By providing access to raw data, this dataset promotes transparency, reproducibility, and further investigations to optimize agricultural practices, paving the way for advancements in crop management.

3. Data Description

The dataset is a collection of images that capture different stages and conditions of lentil crops, including healthy plants and those affected by diseases like ascochyta blight, lentil rust, and powdery mildew. These images were meticulously captured between May and August 2023 at farms in Barishal, Bangladesh, using a Samsung M31 smartphone. Sized at 224 × 224 pixels and stored in JPG format, each image is carefully labeled according to its disease class, making classification and analysis straightforward. Collecting these images was not without its challenges:

- **Time of Day:** Images were captured primarily during the early morning (6 AM to 9 AM) and late afternoon (4 PM to 6 PM) to take advantage of natural lighting, which helps reduce shadows and highlights, ensuring clearer images.
- **Lighting Conditions:** Care was taken to avoid harsh sunlight that can cause glare. The selected times helped maintain consistent lighting across the dataset, minimizing the impact of shadows and enhancing image clarity.

- **Climatic Factors:** The shooting occurred during the late spring to early summer months, characterized by warm temperatures and occasional rain. These conditions allowed for the growth of healthy lentil plants and the observation of various disease symptoms.
- **Stages of Plant Growth Cycle:** Images were collected at different growth stages, including:
  - Seedling stage: Early growth, where initial disease symptoms may start to appear.
  - Vegetative stage: Fully developed leaves, which are essential for photosynthesis.
  - Flowering stage: The transition to reproductive growth, critical for yield.
  - Harvesting stage: Final condition of the plants, where diseases can affect the overall yield.

The dataset successfully includes images across four classes: ascochyta blight, lentil rust, powdery mildew, and normal. Fig. 1 showcases a lentil field from which these images were collected.

The organization of the dataset is presented in Fig 2. The progression of lentil disease development differs based on factors such as the variety of lentil, cultivation conditions, and climatic influences. Disease symptoms can appear and progress at different rates, [4] making timely detection and management crucial for maintaining crop health and yield. Generally, early detection and intervention can prevent widespread damage and ensure optimal crop quality. Table 1 provides a comprehensive explanation of the various disease classes and their symptoms, as well as an explanation of each category in the Lentil Disease Detection Dataset. This classification is vital for the training and validation of machine learning models, which are used to accurately identify and classify lentil diseases. Consequently, it supports effective disease management practices.

This dataset lays the foundation for developing automated disease detection systems, which can significantly enhance the accuracy and efficiency of crop management practices. Automated systems can accurately detect and classify lentil diseases through image analysis. Algorithms that can make accurate recommendations about disease management can be trained using this data via machine learning models. By providing a more sustainable method of crop management, this innovation has the potential to lower labour costs and minimise financial losses for farmers. Controlling quality in lentil crop management can be revolutionised by automating disease detection [5]. Several quality metrics, including disease severity and general crop health, can be assessed by models trained on this dataset. By eliminating human mistake and facilitating inspection and sorting according to strict quality standards, this automation makes it easier to process lentil crops consistently. The result is a more efficient and reliable disease management process that requires less human intervention. This dataset serves as a catalyst for collaboration between computer scientists and agricultural experts. By utilizing this data, researchers can develop and test innovative algorithms, pushing the boundaries of automated disease detection systems. The advancements made possible by this dataset will enhance crop quality and offer significant economic benefits to farmers and the broader agricultural sector. Table 1 provides an overview of the four classes included in the lentil disease dataset. These classes are vital for accurately categorizing the images and training machine learning models to distinguish between healthy and diseased lentil plants.

To facilitate these advancements, this study introduces the lentil disease Dataset. The dataset is organized into two main subfolders: the original dataset comprising images captured directly with a camera, and the augmented dataset containing images generated from the original dataset using data techniques. Fig. 2 illustrates the organizational structure of the dataset.

3.1. Camera specification

The information was collected using the camera of a Samsung M31 smartphone. The Samsung M31 device's camera is equipped with a 64MP Samsung GW1 sensor. The sensor size is 1/1.72 inches, and the individual pixels on the sensor measure 0.8µm each. The camera setup includes an f/1.8 aperture, enabling the capture of high-resolution images under various lighting conditions, which is crucial for accurately identifying disease symptoms in lentil crops. The dataset includes four classes: Ascochyta Blight, Lentil Rust, Powdery Mildew, and Normal. The

Fig. 1. The lentil field from where we collected the dataset images.

Table 1
Details about the lentil disease varieties in the dataset.

Class Name	Description	Visualization
Ascochyta Blight	Ascochyta blight is a fungal disease that affects lentil crops, causing dark lesions on leaves, stems, and pods. This disease can lead to significant yield loss if not detected and managed early. Visible dark spots on leaves and stems, which can coalesce to form larger necrotic areas. Pods may also exhibit similar lesions, impacting seed quality.	
Lentil Rust	Lentil rust is another fungal disease characterized by the presence of small, reddish-brown to orange pustules on the leaves and stems, leading to yellowing and wilting of the plant. This disease can severely affect the photosynthetic ability of the plant, leading to stunted growth and reduced yield. Rust-colored pustules on the undersides of leaves and stems. In severe cases, leaves may turn yellow and drop prematurely.	
Powdery Mildew	Powdery mildew is a fungal disease that covers the leaves and stems with a white, powdery growth. This disease can reduce photosynthesis, leading to weakened plants and lower yields. White, powdery patches on leaves, stems, and pods. Infected areas may become distorted, and severe infections can cause premature leaf drop.	
Normal	This class includes images of healthy lentil plants with no visible disease symptoms. These images are essential for training models to distinguish between healthy and diseased plants. Healthy green leaves, robust stems, and unaffected pods.	

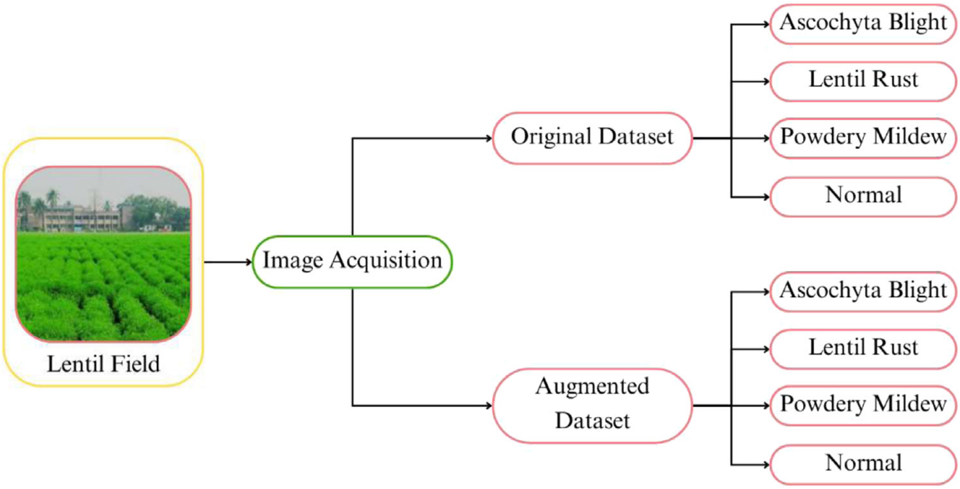

Fig. 2. Organization of lentil disease dataset.

Table 2
Statistics of the lentil disease dataset.

Class Name	Original Images	Augmented Images
Ascochyta Blight	466	1029
Lentil Rust	480	1075
Powdery Mildew	492	1125
Normal	454	1321
Total	1898	4550

original images total 1898, while the data augmentation process has generated an additional 4550 images.

The experimental design focuses on developing a robust dataset for training machine learning models to accurately classify lentil diseases (Table 2).

3.2. Data augmentation

Deep learning algorithms, especially those used for visual object recognition, rely heavily on data augmentation. By generating fresh images from pre-existing ones, improving model generalisation, and decreasing overfitting, it is an effective method for fortifying deep learning models. Shearing, random rotation, horizontal flipping, shifting width and height, zooming, and brightness adjustments were among the augmentation tactics that we employed. Several methods were implemented in line with industry standards to enhance the dataset's diversity and durability. According to these standards, the images can be rotated by up to 40 degrees, among other possible orientations. To further facilitate the bidirectional displacement of the image's information, we included a 0.2 range for both the width and the height shift [6]. A shear range of 0.2 was used to induce the controlled deformation. In order to add additional variety, we adjusted the photo size by using a 0.2 zoom range. Activating horizontal flipping added mirrored copies of the photographs to the dataset. We were able to successfully manage picture alterations by utilising the 'nearest' fill mode. We also adjusted the brightness from half a stop to one and a half to provide a wide range of lighting conditions. These parameter settings enhanced the training of the deep learning models by producing a robust and diverse dataset. A grand total of 5463 augmented photographs were generated from our collection using an automated augmen-

Table 3
Augmented lentil disease dataset distribution.

Class Name	Test	Train	Validation
Ascochyta Blight	111	1029	111
Lentil Rust	95	1075	95
Powdery Mildew	108	1125	108
Normal	143	1321	142
Total	457	4550	456

Table 4
Original lentil disease dataset distribution.

Class Name	Test	Train	Validation
Ascochyta Blight	37	366	37
Lentil Rust	32	380	32
Powdery Mildew	48	392	48
Normal	36	454	36
Total	153	1592	153

tation technique that was code-driven. Our goal in creating these upgraded photographs was to make our dataset more diverse and comprehensive. Table 3 shows the corresponding original sample photographs for each category, expertly linked with these enhanced images. The data augmentation procedure was helpful in expanding and improving our dataset, as shown by this meticulous pairing, which provides a clear and instructive picture of the augmentation findings. We allocated 80% of the dataset to training, 10% to testing, and 10% to validation for our next experiment. A thorough assessment of the model's performance and generalisability is guaranteed by this divide. The model can learn complex patterns of lentil diseases from the training set, which is the main data set. The accuracy of the model is assessed on unknown data by the testing set, while overfitting is prevented during training by fine-tuning it on the validation set. An accurate and dependable system for detecting lentil diseases can be developed with this balanced approach [7].

This dataset for our experiment was also divided into three parts: 80% for training, 10% for testing, and 10% for validation (Table 4).

The training dataset is given controlled variance, which makes the model more adaptable to real-world circumstances. Our main methods include shearing for various viewpoints, [8] horizontal/vertical shifting (up to 20% width/height), and random rotation (0-40 degrees). While horizontal flipping teaches orientation invariance, random zooming (80–120%) aids in managing various scales [9]. A fill mode keeps the image's original content while adjusting the brightness (50–150%) and contrast (70–130%) to account for changes in lighting. Pre-processing adds a random zoom function, increasing the model's versatility. Models are now able to distinguish things in a variety of real-world scenarios thanks to these strategies. Fig. 3 Displays the Augmented images of Lentil crops from the Dataset.

Our deep learning model was tested on the Lentil Disease Dataset using a systematic method that included multiple important steps. Each "node" in a deep learning model is a computational unit, and the model's layers are interconnected. Final predictions are generated by the output layer, which takes data received by the input layer. For activities such as picture categorisation and object detection, the neural network's principal computing capability is in the hidden layers between them. Improving data quality, getting the visual data ready for the model input, and getting the model running efficiently and effectively were all greatly affected by data preparation. This required a number of changes, the first of which was careful data labelling to assign each image to the correct disease class. Image segmentation techniques were applied where necessary to improve dataset quality by removing unwanted background elements. Data preprocessing is fundamental in deep learning as it prepares visual data for model input, en-

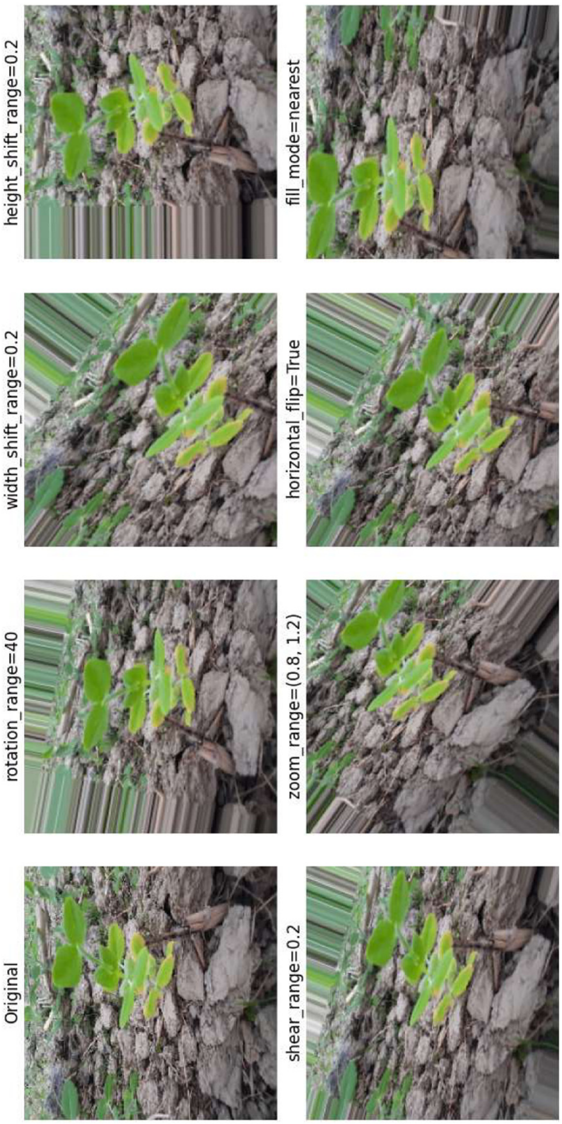

Fig. 3. Random Augmented Image.

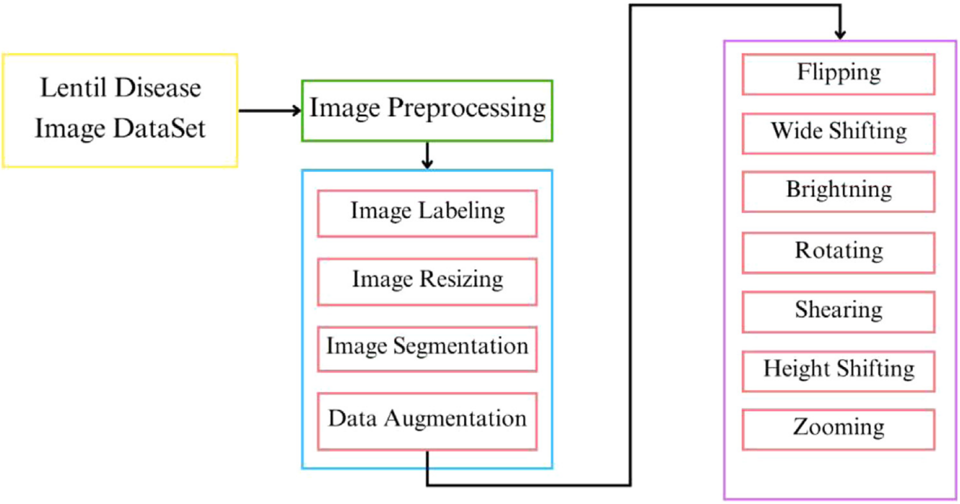

Fig. 4. The pre-processing steps of proposed deep learning model.

hances data quality, and significantly impacts model performance, generalization, and efficiency. Preprocessing involved several critical steps to ensure optimal model training and performance which is shown in Fig. 4.

4.2.1. Data Labeling: Initially, meticulous data labeling was conducted to accurately categorize each image according to its specific lentil disease class (). Precise labeling is essential as it forms the basis for training and refining the DenseNet201 model, enabling it to learn and make reliable predictions.

4.2.2. Image Resizing: Given the variability in image sizes within the dataset, images were resized to a standardized format (e.g., 224×224 pixels). This standardization not only reduced computational demands during training but also ensured consistency in model input, facilitating effective feature extraction by DenseNet201.

4.2.3. Image Segmentation: Where applicable, image cropping techniques were employed to remove irrelevant background elements, thereby enhancing the overall quality and relevance of the dataset for disease classification tasks.

4.2.4. Data Augmentation: Recognizing the need for a robust dataset to enhance model performance and reduce overfitting, various augmentation techniques were applied. These included rotations, shifts, shearing, zooming, and flipping of images. Augmentation effectively expanded the dataset size, introducing diversity and variability crucial for training the DenseNet201 model on a comprehensive range of lentil disease scenarios.

4.2.5. Dataset Splitting: The dataset was meticulously divided into three distinct subsets: training, validation, and testing. The training set, comprising 80% of the data, was used to train the DenseNet201 model on a diverse range of images. The validation set (10%) facilitated ongoing model refinement and parameter tuning, while the independent testing set (10%) served as a final benchmark to evaluate the model's performance on unseen data, ensuring its robustness and reliability [10].

DenseNet201 builds upon the DenseNet architecture by utilizing 201 layers, which include dense blocks and transition layers. The dense connectivity [11] pattern within each dense block facilitates feature reuse and gradient flow, enabling the network to learn discriminative features effectively. This connectivity structure alleviates the vanishing-gradient problem and encourages feature propagation throughout the network, enhancing its ability to learn from small datasets [12]. In DenseNet201, each dense block consists of several convolutional layers, specifically de-

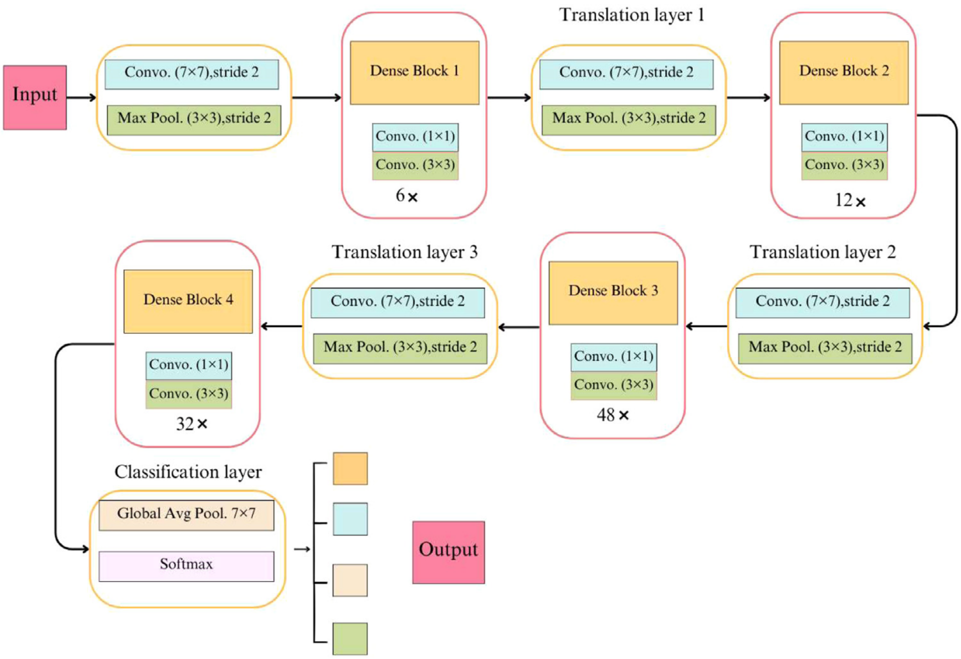

Fig. 5. DenseNet201 model architecture.

signed with varying numbers of filters. For example, the initial layers typically use 64 filters, progressively increasing to 128, 256, and 512 filters in subsequent layers, depending on the depth of the block. Each convolutional layer is followed by batch normalization and ReLU activation functions [13], where batch normalization stabilizes and accelerates the training process by normalizing the input to each layer, while ReLU introduces non-linearity, enabling the model to learn complex representations from the data. Transition layers [14] in DenseNet201 are critical for managing the feature map dimensions. Each transition layer consists of batch normalization, a 1 × 1 convolutional layer for dimensionality reduction, typically reducing the number of feature maps by half, and average pooling to downsample feature maps. These layers reduce the spatial dimensions while preserving essential information, thereby improving computational efficiency and preventing overfitting. The architecture concludes with global average pooling, which aggregates spatial information from the entire feature map, significantly reducing the number of parameters and enhancing generalization. This is followed by a fully connected layer, which typically uses a number of units equal to the number of classes in the classification task. Finally, a softmax activation function is applied to this layer, which outputs probabilities for each class, enabling the model to predict the likelihood of each input image belonging to different disease categories (Fig. 5).

To summarize, the DenseNet201 architecture consists of:

- **Dense Blocks:** Comprising convolutional layers with 64, 128, 256, and 512 filters.
- **Transition Layers:** Including batch normalization, 1 × 1 convolutional layers, and average pooling.
- **Global Average Pooling Layer:** Aggregates features before the final classification layer.
- **Fully Connected Layer:** Outputs predictions for each class based on learned features.
- **Softmax Activation Function:** Produces probability distributions over the classes.

This detailed description of the DenseNet201 model's architecture is essential for understanding its capacity for effective disease detection in lentil crops

4.3.1 Accuracy

Accuracy represents the percentage of correctly predicted images among all predictions, as outlined in Eq. (1). This score evaluates the model's ability to make correct predictions (TP + TN) relative to the total predictions (TP + TN + FP + FN), where TP is true positive, TN is true negative, FP is false positive, and FN is false negative [15].

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

4.3.2 Precision

Precision, denoting [16] the percentage of correctly identified positive outcomes, is calculated using Eq. (2).

Precision (P):

$$P = \frac{TP}{TP + FP} \tag{2}$$

4.3.3. Recall

The recall value [17] assesses the accuracy of positive predictions by comparing true positive findings to the total number of actual positive samples (TP + FN), expressed by Eq. (3).

Recall ⊕:

$$R = \frac{TP}{TP + FN} \tag{3}$$

4.3.4. F1-Score

In evaluating a model's overall performance, [18] researchers often turn to metrics like the F1-score. This score, defined in Eq. (4), captures the harmonic mean of a model's precision and recall.

$$F1 - score \ (F1) :$$

$$F1 = 2 \times \frac{P \times R}{P + R} \tag{4}$$

4.3.5. Confusion Matrix:

Essential for assessing the model's classification performance across multiple classes, the confusion matrix provides a detailed breakdown of predicted versus actual class labels. It aids in identifying model strengths and weaknesses, enabling targeted improvements. Fig. 6 shows the normalize confusion matrix of our without [19] augmented dataset.

The evaluation of the DenseNet201 model [20] on the lentil disease dataset will include these metrics to ensure a comprehensive assessment of its effectiveness in disease classification tasks. The model's ability to generalize to new data and its practical utility in agricultural applications will be highlighted through rigorous testing and validation processes. Table 5 provides a detailed classification report [21,22] for the DenseNet-201 model applied to the lentil disease dataset. The performance metrics included in Table 5 are precision, recall, F1-score [23], and support for each class.

The ROC AUC curve is a crucial metric for evaluating the performance of the DenseNet-201 model in classifying lentil diseases, including Ascochyta blight, Lentil Rust, Normal, and Powdery

Table 5
Classification report for lentil disease detection.

Class Name	Precision	Recall	F1-Score	Support
Ascochyta Blight	0.70	0.84	0.77	74
Lentil Rust	0.86	1.00	0.93	64
Normal	0.94	0.60	0.73	96
Powdery Mildew	0.88	1.00	0.94	72

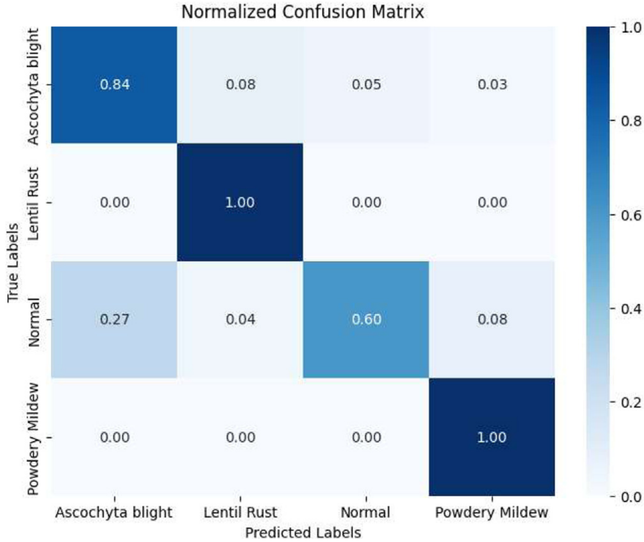

Fig 6. Normalized confusion matrix.

Mildew. The model achieved AUC values ranging across different classes, indicating strong discriminative capabilities, particularly in distinguishing between diseased and healthy plants. The ROC curve also aids in selecting optimal classification thresholds that maximize the True Positive Rate (TPR) while minimizing the False Positive Rate (FPR), enhancing reliability. Additionally, the individual ROC curves highlight the model's strengths and weaknesses, allowing for targeted improvements in classification accuracy. Overall, this analysis demonstrates the DenseNet-201 model's effectiveness in detecting lentil plant diseases, supporting advancements in agricultural practices and plant health monitoring.

4. Future Work

In the future, we aim to leverage advanced deep learning algorithms with image processing to develop a consumer-oriented mobile application. This application will assist in the detection and classification of various lentil diseases, empowering farmers and agricultural professionals with real-time insights and recommendations for effective disease management and crop health.We plan to improve model accuracy by fine-tuning pre-trained networks like EfficientNet and ResNet through transfer learning with domain-specific data augmentation and hyperparameter optimization. Furthermore, we will focus on user-centric app development by designing an intuitive user interface that allows users to easily capture and analyze images of lentil crops. Additionally, we intend to integrate real-time recommendations based on the detected diseases, developing a decision support system that analyzes disease data and provides tailored management practices, such as fungicide application or crop rotation strategies. Lastly, we plan to extend the application to include detection capabilities for other crops and diseases, utilizing the established framework to adapt datasets and algorithms accordingly. By implementing these strategies, we believe the mobile application will significantly enhance farmers' capabilities in managing lentil crop health, ultimately contributing to improved yields and sustainable agricultural practices (Fig. 7).

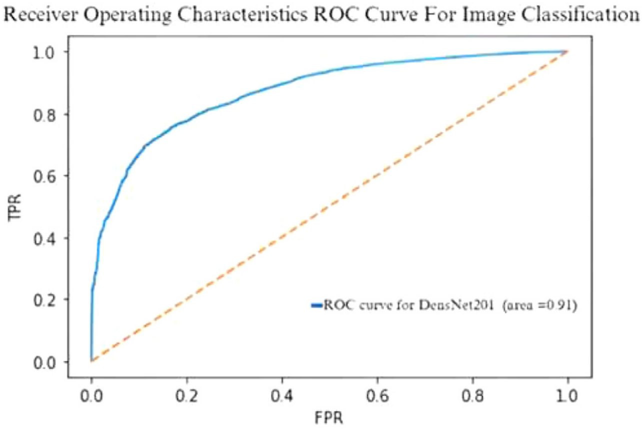

Fig 7. ROC curve for denseNet201 model.

Limitations

The dataset may not be applicable to other parts of the world where lentils are grown because of differences in climate, soil, or agricultural practices, even though it was gathered from several places across Bangladesh. The four-month dataset may fail to encompass all environmental variables, variations in the seasons, and disease progress during a growing season.

Ethics Statement

This article presents a comprehensive dataset designed to enhance the development of deep learning models for disease detection and quality assessment in lentil cultivation. The objective is to improve lentil production efficiency and promote sustainable agricultural practices through technological innovation.

CRediT Author Statement

Eram Mahamud: Conceptualization, Data curation, Methodology, Visualization, Validation, Writing, **Mr. Md Assaduzzaman:** Supervision, Writing – review & editing, Visualization; Shayla Sharmin: Writing – review & editing.

Data Availability

Lentil Plant Disease Image Dataset (4 Class) (Original data) (Mendeley Data).

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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